

JOINT USE OF PHENOTYPIC, PEDIGREE AND GENOMIC INFORMATION IN GENETIC EVALUATION: AN EXAMPLE IN *Eucalyptus grandis*.

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Introduction

Forest tree breeding populations are usually large, with hundreds of progenies from many mother trees. Genotyping costs are thus the main obstacle for the successfully large-scale implementation of genomic selection (GS). A more realistic scenario would include only one sub-set of genotyped trees and fewer genetic markers per sample to obtain the marked-based relationship matrix. Then, the classical pedigree-based relationship matrix could be replaced by the H matrix (e.g. Legarra et al. 2009) that combines two types of genetic information: 1) pedigree information (A matrix) for a group of non-genotyped individuals and 2) marked-based relationship (G matrix) for an additional group of genotyped individuals. This combined approach is appealing as it produces more precise predicted breeding values than those of the pedigree-based counterpart. However, the potential advantage of the combined approach over the classical pedigree-based selection remains uncertain in forest genetic evaluations.

Objectives

- 1) to apply the combined H matrix to estimate (co)variance components in an open-pollinated progeny trial of *Eucalyptus grandis* (Hill ex Maiden);
- 2) to compare the genetic parameters and accuracies of predicted breeding values using the traditional A matrix vs. the combined H matrix;
- 3) to determine the experimental effort required for genotyping different number of trees and their impact on the resulting genetic parameters and accuracies of predicted breeding values;
- 4) to determine if including additional non-genotyped families in the combined H matrix impacts on the results.

Materials and Methods

Table 1. Descriptions of the phenotypic data and families for diameter at breast height (DBH) and total height (HT) for all and genotyped trees in the *Eucalyptus grandis* data set.

	N° of records	Mean (SD)	
		DBH (cm)	HT (m)
Total number of trees	2027	18.85 (4.27)	18.87 (2.68)
Number of trees from families with at least 1 genotyped tree	1637	18.87 (4.24)	18.87 (2.65)
Number of genotyped trees (varied from 1 to 2 per family)	184	20.77 (3.06)	20.56 (1.68)
Total number of families	164	-	-
Number of families with at least 1 genotyped tree	130	-	-

SD = Standard deviation.

- Fifteen SSRs belonging to at least the 11 defined linkage groups of the *Eucalyptus grandis* Genome (Myburg et al. 2014) and showing high PIC (> 0.542) were selected.

- Both univariate and bivariate (for DBH and HT) individual-tree mixed models were studied.

- For the pedigree-based approach, the breeding values vector was assumed distributed as $N \sim (0, A \sigma_a^2)$ where σ_a^2 is the additive genetic variance.

- In the combined approach, the matrix A is substituted by the combined pedigree marker-based pairwise relationship matrix H , whose inverse (H^{-1}) is (e.g. Aguilar et al. 2010):

$$H^{-1} = A^{-1} \begin{bmatrix} 0 & 0 \\ 0 & G^{-1} - A^{-1} \end{bmatrix}$$

- The correction proposed by Powell et al. (2010) was used to overcome the scale problem between G and A_{22} (i.e., the pedigree-based relationship matrix for the genotyped individuals).

Results

Table 2. Variance component and heritability estimates (and approximate standard errors) for diameter at breast height (DBH) and total height (TH) from the different prediction methods studied using the pedigree (A matrix) and the combined approach (H matrix) and the univariate analysis.

Prediction Method	n	Additive variance		Residual Variance		Heritability	
		A	H	A	H	A	H
DBH							
BLUP_A and_H	1637	2.77 (0.84)	3.29 (1.05)	14.47 (0.93)	14.25 (1.02)	0.16 (0.05)	0.19 (0.06)
BLUP_H170	1637	2.77 (0.84)	2.99 (0.97)	14.47 (0.93)	14.47 (0.99)	0.16 (0.05)	0.17 (0.05)
BLUP_H150	1637	2.77 (0.84)	2.79 (0.91)	14.47 (0.93)	14.61 (0.96)	0.16 (0.05)	0.16 (0.05)
BLUP_A142	1788	3.22 (0.86)	3.85 (1.05)	13.97 (0.91)	13.65 (1.00)	0.18 (0.05)	0.22 (0.06)
BLUP_A152	1893	3.39 (0.86)	4.01 (1.03)	13.93 (0.90)	13.62 (0.98)	0.19 (0.05)	0.22 (0.06)
BLUP_A164	2027	3.69 (0.87)	4.27 (1.02)	13.72 (0.89)	13.44 (0.96)	0.21 (0.05)	0.24 (0.05)
TH							
BLUP_A and_H	1637	0.59 (0.25)	0.99 (0.36)	5.63 (0.32)	5.31 (0.36)	0.09 (0.04)	0.14 (0.05)
BLUP_H170	1637	0.59 (0.25)	0.86 (0.33)	5.63 (0.32)	5.42 (0.35)	0.09 (0.04)	0.12 (0.05)
BLUP_H150	1637	0.59 (0.25)	0.71 (0.29)	5.63 (0.32)	5.55 (0.33)	0.09 (0.04)	0.10 (0.04)
BLUP_A142	1788	0.67 (0.25)	1.07 (0.35)	5.58 (0.31)	5.25 (0.35)	0.10 (0.04)	0.15 (0.05)
BLUP_A152	1893	0.71 (0.25)	1.09 (0.34)	5.56 (0.30)	5.24 (0.34)	0.10 (0.04)	0.15 (0.05)
BLUP_A164	2027	0.69 (0.24)	1.04 (0.32)	5.68 (0.30)	5.41 (0.33)	0.10 (0.04)	0.14 (0.04)

Figure 1. Increase in the accuracy of breeding values (absolute values and percentages) from the combined approach (H matrix) compared to the classical pedigree (A matrix) for non-genotyped parents and offspring and genotyped offspring, and for diameter at breast height (DBH) and total height (TH) using the univariate and bivariate analyses.

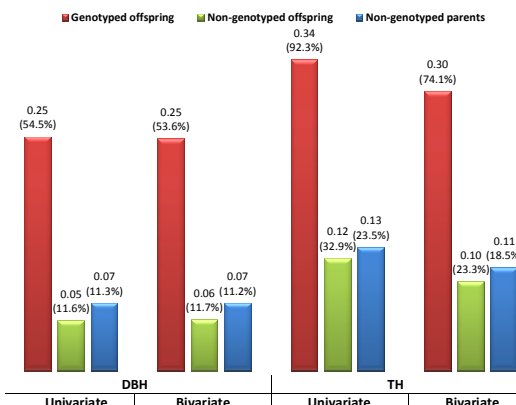


Figure 2. Percent increase in the accuracy of breeding values from the combined approach (H matrix) compared to the classical pedigree (A matrix) vs. the number of genotyped offspring per mother for parents and offspring, and for diameter at breast height (DBH) and total height (TH) in different prediction methods studied using the univariate and bivariate analyses.

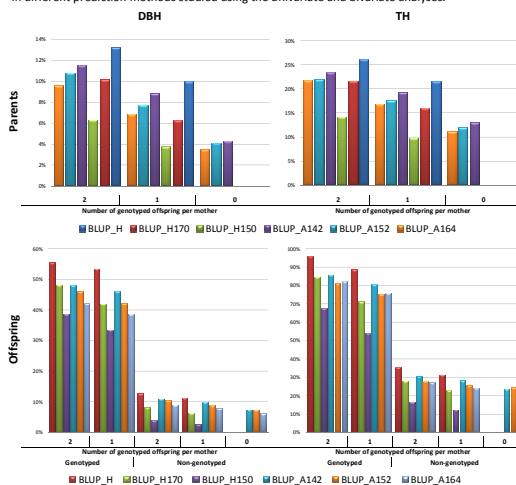


Figure 3. Percent decrease in the accuracy of breeding values for mothers from the combined approach (H matrix) compared to the classical pedigree (A matrix) vs. the number of offspring per mother for diameter at breast height (DBH) and total height (TH).

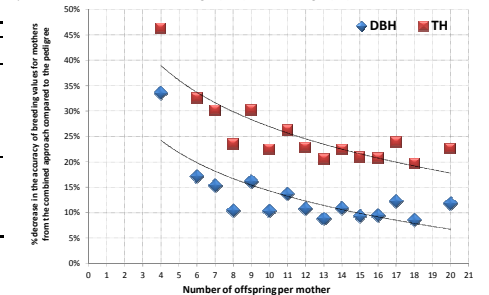
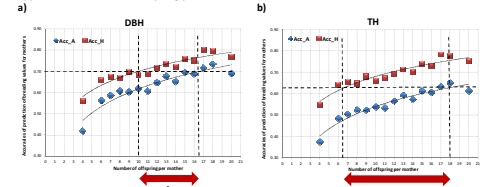


Figure 4. Accuracy of breeding values for mothers from the pedigree (A matrix) and the combined approach (H matrix) vs. the number of offspring per mother for: a) diameter at breast height (DBH) and b) total height (TH). Horizontal dash line shows an average accuracy of breeding value for mothers of 0.70 (DBH) and 0.62 (TH). The 2 vertical lines show the approximate number of offspring per mother.



Conclusions

- Combined approach achieved higher accuracies probably because genomic information improved prediction of the Mendelian sampling terms.
- The increase in the accuracy was more evident in the trait with smaller heritability (TH).
- Results from the bivariate analysis are similar to those from the univariate analyses.
- When decreases the number of genotyped trees, or increases the number of families with non-genotyped trees, the accuracy of BVs decreased.
- When increases the number of offspring per mother, differences in accuracy of mother BVs from the A and H matrices decreased for both traits studied. This decrease was more evident in the trait with smaller heritability (TH).
- The increase in the accuracy of mother BVs was the equivalent of having data from 6 (DBH) and 11 (TH) additional progenies for each mother.
- Implementation of the combined approach is straightforward (we can use standard software).
- Combined approach utilized for the current generation should be tested in the subsequent generation.

References

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- Myburg AA et al. (2014). Nature 510: 356-362.
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